

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Debugging

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COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO2: Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete. (BTL6)

Assessment Tool Weightage: 25%

Question Count: 10 **Total Marks: 20**

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Debugging

Criteria	Problem Identification (4 Marks)	Debugging (4 Marks)	Optimization (4 Marks)	Analysis (4 Marks)	Code Quality (4 Marks)
Marks					

**Total
Marks Awarded:**

Questions:

Question 1: SQLite Database Connection

The following Python program is written to connect to an SQLite database and create a table. However, the program raises an error and does not execute successfully. Analyze the code, identify the mistakes, and rewrite the corrected version.

```
import sqlite3
conn = sqlite3.connect(student.db)
cursor = conn.cursor
cursor.execute("""
CREATE TABLE Student(
```

```

Student_ID INTEGER PRIMARY KEY,
Name TEXT,
Department TEXT
)
"""
conn.commit
conn.close()

```

CORRECT CODE:

```

import sqlite3

conn = sqlite3.connect("student.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE Student(
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Department TEXT
)
""")

conn.commit()
conn.close()

```

Question 2: MySQL Record Insertion

The following Python program is intended to connect to a MySQL database and insert a new student record. The program contains multiple errors that prevent successful execution. Debug the code and provide the corrected program.

```

import mysql.connector
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
    database="college"
)
cursor = conn.cursor()
query = "INSERT INTO Student(Student_ID, Name, Department) VALUES
(%s,%s,%s)"
cursor.execute(query, (101, "Alice"))
conn.commit()
conn.close()

```

CORRECT CODE:

```

import mysql.connector

```

```

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
    database="college"
)

cursor = conn.cursor()

query = "INSERT INTO Student(Student_ID, Name, Department) VALUES
(%s, %s, %s)"
cursor.execute(query, (101, "Alice", "CSE"))

conn.commit()
conn.close()

```

Question 3: PostgreSQL Data Retrieval

The following Python program is designed to retrieve all records from the Employee table in a PostgreSQL database. The code contains logical and syntax errors. Analyze the program, identify the errors, and write the corrected version.

```

import psycopg2
conn = psycopg2.connect(
    host=&quot;localhost&quot;,
    database=&quot;company&quot;,
    user=&quot;postgres&quot;,
    password=&quot;admin&quot;
)
cursor = conn.cursor()
cursor.execute(&quot;SELECT * FROM Employee&quot;)
rows = cursor.fetchone()
for row in rows:

    print(row)
cursor.close()

```

CORRECT CODE:

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="admin"
)

```

```

cursor = conn.cursor()

cursor.execute("SELECT * FROM Employee")
rows = cursor.fetchall()

for row in rows:
    print(row)

cursor.close()
conn.close()

```

Question 4: SQLite Record Update

The following program attempts to update the department of a student in an SQLite database. The update operation fails due to errors in the code. Identify the mistakes and rewrite the corrected program.

```

import sqlite3
conn = sqlite3.connect("&quot;college.db&quot;")
cursor = conn.cursor()
query = &quot;UPDATE Student SET Department=&#39;ECE&#39; WHERE
Student_ID=?&quot;
cursor.execute(query)
conn.commit()
conn.close()

```

CORRECT CODE:

```

import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

query = "UPDATE Student SET Department=? WHERE Student_ID=?"
cursor.execute(query, ("ECE", 101))

conn.commit()
conn.close()

```

Question 5: PostgreSQL Record Deletion

The following Python program is intended to delete a customer record from a PostgreSQL database. The code contains errors related to SQL execution and resource management. Debug the program and provide the corrected version.

```

import psycopg2
conn = psycopg2.connect(

```

```

host=&quot;localhost&quot;,
database=&quot;bank&quot;,
user=&quot;postgres&quot;,
password=&quot;admin&quot;
)
cursor = conn.cursor()
customer_id = 105
cursor.execute(&quot;DELETE FROM Customer WHERE
Customer_ID=%s&quot;)
conn.close()

```

CORRECT CODE:

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="admin"
)

cursor = conn.cursor()

customer_id = 105
cursor.execute("DELETE FROM Customer WHERE Customer_ID=%s",
(customer_id,))

conn.commit()

cursor.close()
conn.close()

```

Question 6: SQLite Record Retrieval

The following Python program is intended to retrieve the details of a student whose Student_ID is 105 from an SQLite database. The program produces an error and does not display the required record.

Analyze the code, identify the mistakes, and rewrite the corrected version.

```

import sqlite3
conn = sqlite3.connect(&quot;college.db&quot;)

cursor = conn.cursor()
student_id = 105
cursor.execute(&quot;SELECT * FROM Student WHERE
Student_ID=?&quot;, student_id)
row = cursor.fetchall()
print(row[0])

```

```
conn.close()
```

CORRECT CODE:

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

student_id = 105
cursor.execute("SELECT * FROM Student WHERE Student_ID=?", (student_id,))

row = cursor.fetchone()

print(row)

conn.close()
```

Question 7: MySQL Table Creation

A student wrote the following Python program to create a table in a MySQL database. The program contains syntax and logical errors. Debug the code and provide the corrected version.

```
import mysql.connector
conn = mysql.connector.connect(
    host=&quot;localhost&quot;,
    user=&quot;root&quot;,
    password=&quot;root&quot;,
    database=&quot;college&quot;
)
cursor = conn.cursor()
cursor.execute(&quot;&quot;&quot;
CREATE TABLE Student(
Student_ID INT PRIMARY KEY
Name VARCHAR(50),
Department VARCHAR(30)
)
&quot;&quot;&quot;)
conn.commit()
conn.close()
```

CORRECT CODE:

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
```

```

        database="college"
    )

    cursor = conn.cursor()

    cursor.execute("""
CREATE TABLE Student(
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Department VARCHAR(30)
)
""")

    conn.commit()
    conn.close()

```

Question 8: PostgreSQL Record Insertion

The following Python program is intended to insert a new employee record into a PostgreSQL database.

The program fails during execution due to multiple errors. Analyze the code, identify the mistakes, and rewrite the corrected version.

```

import psycopg2
conn = psycopg2.connect(
    host=&quot;localhost&quot;,
    database=&quot;company&quot;,
    user=&quot;postgres&quot;,
    password=&quot;admin&quot;
)
cursor = conn.cursor()

sql = &quot;&quot;&quot;
INSERT INTO Employee(Employee_ID, Name, Salary)
VALUES (%s, %s, %s)
&quot;&quot;&quot;
cursor.execute(sql, 101, &quot;John&quot;, 55000)
conn.commit()
cursor.close()
conn.close()

```

CORRECT CODE:

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="company",
    user="postgres",
    password="admin"
)

```

```

cursor = conn.cursor()

sql = """
INSERT INTO Employee(Employee_ID, Name, Salary)
VALUES (%s, %s, %s)
"""

cursor.execute(sql, (101, "John", 55000))

conn.commit()

cursor.close()
conn.close()

```

Question 9: MySQL Record Update

The following Python program is intended to update the salary of an employee in a MySQL database. The update operation does not execute successfully. Debug the code and provide the corrected version.

```

import mysql.connector
conn = mysql.connector.connect(
    host=&quot;localhost&quot;,
    user=&quot;root&quot;,
    password=&quot;root&quot;,
    database=&quot;company&quot;
)
cursor = conn.cursor()
salary = 65000
emp_id = 110
query = &quot;UPDATE Employee SET Salary=%s WHERE
Employee_ID=%s&quot;
cursor.execute(query, salary, emp_id)
conn.commit()
print(&quot;Record Updated&quot;)
cursor.close()
conn.close()

```

CORRECT CODE:

```

import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="root",
    database="company"
)

```



```

cursor = conn.cursor()

salary = 65000
emp_id = 110

query = "UPDATE Employee SET Salary=%s WHERE Employee_ID=%s"
cursor.execute(query, (salary, emp_id))

conn.commit()

print("Record Updated")

cursor.close()
conn.close()

```

Question 10: PostgreSQL Record Deletion

The following Python program attempts to delete all customer records belonging to the city Chennai.

However, the program contains errors related to parameter passing and transaction handling. Analyze the code, identify the mistakes, and rewrite the corrected version.

```

import psycopg2
conn = psycopg2.connect(
    host=&quot;localhost&quot;,
    database=&quot;bank&quot;,
    user=&quot;postgres&quot;,

    password=&quot;admin&quot;
)
cursor = conn.cursor()
city = &quot;Chennai&quot;
query = &quot;DELETE FROM Customer WHERE City=%s&quot;
cursor.execute(query)
print(&quot;Records Deleted&quot;)
cursor.close()
conn.close()

```

CORRECT CODE:

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="bank",
    user="postgres",
    password="admin"
)

cursor = conn.cursor()

```

```
city = "Chennai"
```

```
query = "DELETE FROM Customer WHERE City=%s"  
cursor.execute(query, (city,))
```

```
conn.commit()
```

```
print("Records Deleted")
```

```
cursor.close()  
conn.close()
```